

processor, which is setting a new performance per watt benchmark. It finds its sweetest spot in 15W TDP designs. Compared to modules with V1608B processors, conga-TCV2 deliver a perfor-

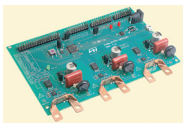


mance of 97% (V2516) to 140% (V2718) with up to 8 cores. Due to the 7nm Zen 2 cores, single-core performance also increased between 25% and 35%, making performance improvements in 24/7 connected and fanless embedded systems. Applications include multi-functional industrial edge gateways, digital signage systems, gaming terminals, and infotainment platforms. Multi-headed medical imaging systems in the operating theatre, machine vision, and machine learning systems are further target markets.

congatec
www.congatec.com

Energy meter evaluation board

ST's EVALSTPM-3PHISO evaluation board speeds design of 3-phase AC watt meters using low-cost, electromagnetic-immune shunt sensors, and advanced galvanic-isolation technology for superior reliability and robustness. It combines



the high accuracy STPMS2 metering front-end IC and the advanced STISO621 digital isolator with

customisable turnkey firmware running on an STM32 microcontroller to compute metrology and power-quality data. The sensing circuitry and PCB layout are optimised to ensure robustness against EMI and a strong signal-to-noise ratio for high-accuracy measurement and post-processing computation. The board can also be used as a reference for multi-phase industrial metering applications beyond utility metering, such as electric-vehicle charging, servers, and solar inverters.

STMicroelectronics
www.st.com

2nd Gen 5G NR modules

Quectel's second-generation 5G new radio (NR) modules are based on the latest Qualcomm Snapdragon X65 and X62 5G modem-RF systems and satisfy the needs of industries that require

enhanced mobile broadband and reliable communication capabilities. The modules will speed up commercialisation of 5G technologies and bring the benefits of mobile broadband to fixed



wireless access, industrial IoT, mobile compute, telemedicine, private networks, and more. The new sub-6GHz modules include the RG520F (LGA form factor), the RM520F (M.2) based on Snapdragon X65, the RG520N (LGA), and the RM520N (M.2) based on Snapdragon X62. The modules support all three combinations of sub-6GHz time division duplex (TDD) and frequency division duplex (FDD) carrier aggregation (CA), including CA of FDD + TDD, FDD + FDD, and TDD + TDD. This ensures greater 5G coverage, capacity, and throughput.

Quectel
www.quectel.com

Radiation-hardened power converters

Microchip has expanded its SA50-120 power converter family with nine new units based on commercial off-the-shelf technology. The radiation-hardened DC to DC power converters utilise



surface mount component construction, allowing flexibility and customisation to fit specific

requirements and enable designers to quickly scale up development, reducing risk and development time. SA50-120 utilise 120V input and offer up to 56W of output in a low-profile, small size. They offer eight million hours mean time between failure and up to 87% efficiency and are qualified to 100krad (Si) total ionisation dose and single event effects greater than 80MeV cm²/mg.

Microchip Technology
www.microchip.com

T&M SOLUTIONS

Source measure units

R&S NGU201 and NGU401 source measure units (SMUs) enable simultaneous sourcing and precise measurement of currents and voltages. The two-quadrant NGU201 addresses wireless

device battery tests and automatically switches from source mode to sink mode at a defined positive input voltage. The four-quadrant NGU401 can also switch at negative voltages, supporting source measurements for a vast range of power supply types. To handle rapidly changing load conditions typical of battery-powered communication devices switching between sleep and transmit modes, the NGU201 features optimised control



loop load, providing recovery time of under 30µs with minimum overshoot. It includes six current ranges from 10µA with 100pA resolution to 10A with 10µA resolution. NGU401 is the specialist for semiconductor testing. It provides source measurements for any equipment requiring source and sink voltages in the range of -20V to +20V. In addition, it includes a fast current regulation mode to avoid damaging current-sensitive devices under test, such as LEDs or other semiconductors.

Rohde & Schwarz
www.rohde-schwarz.com

Integrated measurement software

IS8000 integrated measurement software platform from Yokogawa controls and collects data and generates reports from multiple measuring instruments. It also supports connection to such devices as



high-speed cameras from other companies. The software can aid in the development of motors, inverters, and other types of equipment by synchronising the measurement data from power analysers with the multichannel analogue measurement signal data from a high-speed data logger. As a total solution that will help develop products more efficiently, this software will be offered together with other Yokogawa products such as power analysers and the DL950 ScopeCorder.

Yokogawa Test and Measurement
www.tmi.yokogawa.com

Integrated instruments with GUI

Keysight portfolio of smart essential lab bench products includes a triple-output